TRIVIAL RATIONAL CONTAMINATION IN PSLQ-BASED POLYNOMIAL CONTINUED FRACTION SEARCHES: DIAGNOSIS, CORRECTION, AND A PRE-SCREENING PROTOCOL

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Abstract. We identify two mechanisms that produce rational limits in random polynomial continued fractions (PCFs), both capable of generating false positives in PSLQ-based searches for transcendental identities. First, if $a(1) = 0$ then the GCF $b(0) + \mathbf{K}(a(n)/b(n))$ reduces trivially to $b(0)$ (the first partial numerator vanishes). Second, if $a(k) = 0$ for some $k \geq 2$, the CF terminates at finite depth. Together these account for *all* 43 rational limits in a 1000-family survey across degrees 2–6. We propose a pre-screening protocol that eliminates both classes before extended-basis PSLQ, and document the five-iteration discovery arc—including a Wallis initialization bug that temporarily masked the triviality of the first mechanism—as a case study in machine-assisted self-correction.

1. Introduction

A *polynomial continued fraction* (PCF) is a generalized continued fraction

$$K = b(0) + \cfrac{a(1)}{b(1) + \cfrac{a(2)}{b(2) + \cfrac{a(3)}{\ddots}}} \tag{1}$$

where $a(n) = \sum_{i=0}^{d} \alpha_i \, n^i$ and $b(n) = \sum_{i=0}^{d} \beta_i \, n^i$ are polynomials of degree d with integer coefficients. Following the Ramanujan Machine project [1], random sampling of such families has become a standard tool for discovering new identities involving mathematical constants.

Any PSLQ-based search for transcendental identities among PCF limits must contend with a practical nuisance: a non-negligible fraction of random families converge to *rational* values, which can trigger PSLQ relations that

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mimic genuine transcendental identities when the basis includes redundant constants. In a survey of 1000 random PCF families (200 per degree $d = 2, \ldots, 6$, coefficients in $\{-4, \ldots, 4\}$), we found 43 such families—initially misidentified as evidence for $\ln 2$ universality.

The correct explanation is elementary:

Theorem 1 (Trivial rational limit)**.** *Let $a(n), b(n) \in \mathbb{Z}[n]$. If $a(1) = 0$, then $K = b(0)$.*

Proof. The first partial numerator $a(1)$ is the numerator of the entire tail $a(1)/(b(1) + \cdots)$. When $a(1) = 0$, the tail vanishes and $K = b(0) + 0 = b(0)$. □

The condition $a(1) = 0$ means $\sum_{i=0}^{d} \alpha_i = 0$—the sum of the coefficients of the numerator polynomial vanishes. This is a linear constraint on the $d + 1$ coefficients and is satisfied by $\sim$5–9% of random families depending on degree (Section 3.3). Although elementary, this criterion identifies a non-negligible fraction of random PCF families as trivially rational—sufficient to dominate any PSLQ search that does not filter them and to produce false-positive transcendental identity claims. We state it as a theorem rather than a remark precisely because its role is *diagnostic*: it is the first step in the pre-screening protocol of Section 4.

Proposition 2 (Finite CF)**.** *If $a(k) = 0$ for some integer $k \geq 2$ and $a(n) \neq 0$ for $1 \leq n < k$, then the GCF(1) terminates at depth $k - 1$ and is rational.*

Proof. The k-th partial numerator vanishes, so the tail from depth k onward contributes zero. The result is a depth-$(k-1)$ finite continued fraction, which is a rational function of its entries. □

Together, Theorem 1 and Proposition 2 explain all 43 rational limits in our survey (Table 1), with zero residual cases.

2. The Wallis recurrence and a convention warning

The standard Wallis recurrence for the convergents of (1) is

$$A_{-1} = 1, \quad A_0 = b(0), \qquad B_{-1} = 0, \quad B_0 = 1, \tag{2}$$

with

$$A_n = b(n)\, A_{n-1} + a(n)\, A_{n-2}, \qquad B_n = b(n)\, B_{n-1} + a(n)\, B_{n-2}, \tag{3}$$

for $n \geq 1$. The n-th convergent is $K_n = A_n/B_n$.

An alternative implementation computes the *tail* CF $T = a(1)/(b(1) + \cdots)$ separately and returns $K = b(0) + T$. For the tail, the correct Wallis initialization is $P_0 = a(1)$, $Q_0 = b(1)$, giving $T = P_n/Q_n$.

Remark 3 (The initialization bug). Our original evaluator used the incorrect initialization

$$\texttt{Pc} = b(1), \quad \texttt{Qc} = 1 \qquad \text{(wrong)}$$

instead of the correct

$$\texttt{Pc} = b(0), \quad \texttt{Qc} = 1, \quad \text{with } K_n = A_n/B_n \qquad \text{(standard Wallis)}$$

and returned $b(0)+\texttt{Pc}/\texttt{Qc}$. This computes $b(0)+[b(1)+T]$ instead of $b(0)+T$, adding an offset of exactly $b(1)$ to every PCF value.

Concrete example. Let $a(n) = n$ and $b(n) = 2 + n$ (so $a(1) = 1$, $b(0) = 2$, $b(1) = 3$). The CF is $2 + 1/(3 + 2/(4 + 3/(5 + \cdots)))$. The standard Wallis gives $K \approx 2.2956$; the bugged evaluator returns $2.2956 + 3 = 5.2956$.

The error is invisible to convergence testing (the CF still converges, just to the wrong value) and is not caught by PSLQ rationality checks (a shifted rational is still rational). It was detected only when the "theorem" $K = b(0) + b(1)$ was subjected to a convention audit against the golden ratio $\varphi = 1 + 1/(1 + 1/(1 + \cdots))$ (Section 5, Iteration 11). The bug has been patched; the corrected evaluator and the patch commit are recorded in the project repository.[1]

We emphasize this because the bug is easy to introduce and hard to detect: any code that splits the CF into $b(0)$ plus a tail must take care with the tail's initial conditions. The standard formulation (2)–(3) avoids the issue entirely.

3. Empirical classification

3.1. **Survey design.** For each degree $d \in \{2, 3, 4, 5, 6\}$, we generated 200 random PCF families with coefficients α_i, β_i drawn uniformly from $\{-4, \ldots, 4\}$ (leading coefficients and β_0 forced nonzero). Each family was evaluated using the standard Wallis recurrence (2)–(3) to depth 2000 at 150-digit precision, and the limit K was tested for rationality via PSLQ on the basis $[1, K]$.[2]

Table 1. Classification of the 43 rational PCF limits (corrected evaluation).

Mechanism	**Condition**	**Count**	**Limit**
Trivial zero (Thm. 1)	$a(1) = 0$	34	$K = b(0)$
Finite CF (Prop. 2)	$a(k) = 0,\ k \geq 2$	9	$K \in \mathbb{Q}$
Non-trivial rational	—	0	—
Total		**43/43**	

3.2. **Results.** All 34 trivial-zero cases satisfy $K = b(0)$ exactly, verified by exact rational arithmetic (`fractions.Fraction`) and by backward evaluation from depth 100.

[1]Bug patch: `siarc_t1t6_relay.py`, function `eval_pcf_forward`. All results in this paper use the corrected standard Wallis (2)–(3).

[2]Reproduce: `python scripts/iter11_bugfix_reeval.py`. Full code at `https://github.com/papanokechi/pcf-rational-contamination`.

The 9 finite CF families terminate at $a(k) = 0$ for $k \in \{2, 3\}$, yielding distinct rational values: $-29/13$, $-12/5$, $-7/4$, -1, $2/3$, $8/7$, $3/2$, $28/9$, 4. Each value was verified by exact rational arithmetic (`fractions.Fraction`) at the termination depth.

No family produced a non-trivial rational limit (i.e., $K \in \mathbb{Q}$ with $K \neq b(0)$ and no finite termination). No family that was rational under the bugged evaluator became irrational after correction.

3.3. **Density model.** The probability that a random family has $a(1) = 0$ is $\Pr\bigl(\sum_{i=0}^{d} \alpha_i = 0\bigr)$, where each α_i is uniform on $\{-4, \ldots, 4\}$ (the leading coefficient restricted to $\{-4, \ldots, 4\} \setminus \{0\}$). The leading coefficient α_d is restricted to $\{-4, \ldots, 4\} \setminus \{0\}$, contributing variance $15/2$ rather than the $20/3$ of the remaining coefficients. By a CLT approximation with total variance $\sigma^2 = d \cdot \frac{20}{3} + \frac{15}{2}$,

$$\Pr(a(1) = 0) \;\approx\; \frac{1}{\sqrt{2\pi\sigma^2}},$$

yielding $\approx 8.7\%$ at $d = 2$ and $\approx 5.8\%$ at $d = 6$. Empirically: 7.5% ($d = 2$), 6.5% ($d = 3$), 8.5% ($d = 4$), 8.5% ($d = 5$), 4.5% ($d = 6$), consistent within sampling noise. The finite CF mechanism adds $\sim$1–2% to the total contamination rate.

4. Pre-screening protocol

Based on the analysis above, we recommend the following pre-screening steps before running extended-basis PSLQ on a PCF family:

(1) **Evaluate $a(1)$.** If $a(1) = \sum \alpha_i = 0$, the family is trivially rational ($K = b(0)$). Discard.
(2) **Check for integer roots of a.** If $a(k) = 0$ for any $k \geq 2$, the CF terminates at finite depth. Compute the finite CF value; if rational, discard or flag as known.
(3) **Rationality pre-check.** Before enriched-basis PSLQ (e.g., basis $[1, K, \ln 2, K \ln 2, (\ln 2)^2]$), first test the minimal basis $[1, K]$. If PSLQ returns a relation, K is likely rational and will contaminate any enriched scan.
(4) **Convention audit.** Verify the CF evaluator against a known identity (e.g., $1 + 1/(1 + 1/(1 + \cdots)) = \varphi$) before trusting numerical values.

Steps 1–2 are $O(d)$ coefficient checks requiring no high-precision arithmetic. Step 3 adds a single PSLQ call per family. Together they eliminated 100% of false positives in our survey.[3]

[3]The pre-screen is implemented in `scripts/ln2_prefilter.py`.

5. Discovery narrative

The results above were reached through five iterations of an automated conjecture refinement relay (AEAL architecture [3, 4]), providing a documented case study of machine-assisted self-correction:

(1) **Iteration 7** (false positive): A PSLQ scan with the enriched basis $[1, K, \ln 2, K{\cdot}\ln 2, (\ln 2)^2]$ found 43 hits, initially claimed as evidence that $\ln 2$ appears in PCF limits at positive density across all degrees.
(2) **Iteration 8** (rationality diagnosis): All 43 relations have the form $[a, b, 0, 0, 0]$ or factor as $(c_1K+c_0)(1+\ln 2) = 0$, both implying $K \in \mathbb{Q}$. Zero genuine $\ln 2$ identities. The 44 associated claims were retracted.
(3) **Iteration 9** (partial model): Seven families with $K{=}{-3}$ were characterized; all share $b(0) + b(1) = -3$ with $b(0) \in \{-4, -3, -2, -1\}$. A "dynamical attractor" hypothesis was proposed and later superseded.
(4) **Iteration 10** (apparent theorem): The pattern $a(1) = \sum \alpha_i = 0$ was identified across all 34 non-finite families. A Wallis induction proof was constructed for the claim "$K = b(0) + b(1)$." This was accepted as a theorem.
(5) **Iteration 11** (bug discovery): A convention audit (Remark 3) revealed that the evaluator adds $+b(1)$ to every PCF value. The "theorem" $K = b(0) + b(1)$ was a documentation of the code bug. The corrected result is the trivial $K = b(0)$. The Wallis induction proof was valid *within the bugged recurrence* but does not describe the standard mathematical continued fraction.

The full arc—false positive → rationality detection → attractor hypothesis → apparent theorem → bug discovery → trivial explanation—is, to our knowledge, the first documented instance of an automated relay generating a false positive, proposing and then retracting an intermediate "theorem," and ultimately converging on the correct (trivial) result through self-correction. To our knowledge, this is a documented instance of such a complete arc within an automated relay. All iterations, including retractions, are preserved in the governance log.[4]

6. Discussion

Why this matters. The contamination rate of ∼5–9% is high enough to produce false positives in any PSLQ scan of moderate size. In our survey, 43 out of 1000 families (4.3%) were affected. For larger coefficient ranges or higher degrees, the rate changes predictably via the density model, allowing experimenters to estimate their expected false-positive load *a priori*.

The value of documenting errors. The Wallis initialization bug is a cautionary example. It produces a systematic offset that is invisible to convergence diagnostics and rationality checks, yet transforms a trivial

[4]AEAL claims log: `results/claims.jsonl`. Convention audit: `scripts/iter11_bugfix_reeval.py`.

observation ($a(1) = 0 \Rightarrow K = b(0)$) into an apparently non-trivial theorem ($K = b(0) + b(1)$) that survives induction proof—because the proof is valid for the bugged recurrence. The only reliable diagnostic was cross-checking against a known identity with an independent evaluator. We recommend this as standard practice for any computational continued fraction research.

Relation to known results. The observation that $a(1) = 0$ makes the GCF trivial is implicit in the definition—the first partial numerator is zero, so the tail contributes nothing—but does not appear as an explicit warning in the standard references [2], likely because analytic convergence theory assumes $a(n) \neq 0$ for all n. In the context of random polynomial PCFs, where $a(1) = 0$ occurs at positive density, the observation becomes practically relevant.

Broader impact. The contamination mechanism identified here—overcomplete PSLQ bases combined with unvetted evaluator initialization—is not specific to our relay. Any system that runs PSLQ on programmatically generated continued fractions faces the same risk, including the Ramanujan Machine [1] and its derivatives. The pre-screening protocol of Section 4 is directly applicable: steps 1–2 require only coefficient arithmetic and can be added to any PCF search pipeline at negligible cost. We hope that documenting both the contamination mechanisms and the process by which they were discovered—including the intermediate errors—provides a useful template for quality control in automated mathematical discovery.

References

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